In the last chapter, we created tables of the Favorita Sales Forecasting dataset from Kaggle. Now it's time to explore! Open up the first notebook, `CH4-01-Exploring Favorita Sales Data`, to do some initial data exploration in SQL:

1. The first cell is a simple `select *` SQL command. After running the cell, focus on the in-cell **Data Profile** and **Visualization** options.

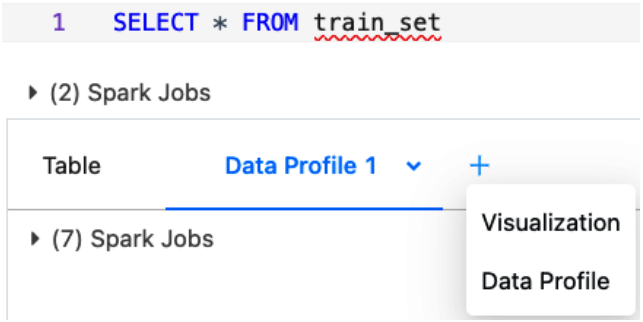


Figure 4.16 – You can create visualizations and a data profile of a SQL query result in a notebook

2. Choose **Data Profile** and investigate the information autogenerated about the data.
3. Choose **Visualization**. You can create a line chart with the date (month) versus the sum of sales and grouped by family. Notice that not all data is used to produce the visualization; we only see January at first. Once you save the visualization, the chart will display in the cell. The **Truncated data** message should be present at the bottom of the visualization. To increase the number of records in the chart, select **Aggregate over more data**.
4. Continue to play with the options. In the next cell, we filter to the top-performing product families. Investigate how different or similar the charts appear.

Now that we've done some initial exploration of the Favorita data, we can run a Databricks AutoML experiment to generate a baseline model. AutoML can be launched in the UI, which we demonstrated earlier in this chapter in the *Generating data profiles with AutoML* section (*Figure 4.11*), or you can create an experiment via an API as shown in *Figure 4.17*. For this project, we will launch both a regression experiment and a forecasting experiment. Let's start with the regression run.

Notice that at the top of the notebook, the default language is SQL rather than Python. Therefore, when we want to execute Python code, we need to include `%python` at the top of the cell. We use Python for AutoML in the last cell of the first notebook. We've set the `timeout_minutes` variable to `30`, so the experiment will run for up to 30 minutes. However, AutoML stops training models if the validation metric is no longer improving. In that case, the experiment will finish in less time. Once the run is complete, the notebook UI will display links to the MLflow experiment where the model versions are accessible, the best trial notebook with the best model's code, and the data exploration notebook. Since this chapter focuses on exploring data, we will only open the data exploration notebook for now.